"On my honor I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to horsh discipline"

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Question 1-

$$A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}.$$

a)

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Question 2-

a-) $x - 5y = 0 \rightarrow$ max rank $= 1$

$N(A) \Rightarrow K -$ rank $\Rightarrow$ Dimension inside R^K

if $K - r \rightarrow$ max rank $= 1$ $\quad K - 1 = 3 \rightarrow$ to have value

$\boxed{K = 4}$

minimum

b) $x - 5y = 0$, $x = 0$ this is my expression in R^4 (4 = K)

$$\begin{bmatrix} 1 & -5 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-R_1 + R_2 \rightarrow R_2$

$$\begin{bmatrix} 1 & -5 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

rank $= 1$

$n - r = 4 - 1 = 3D$ inside R^4

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Question 2- (c)

$x = 5y$

$x - 5y = 0$

$$\begin{bmatrix} 1 & -5 & 0 \\ 1 & 4 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

$-R_1 + R_2 \Rightarrow R_2$

$$\begin{bmatrix} 1 & -5 & 0 \\ 0 & 9 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

$\frac{1}{9} \cdot R_2 \Rightarrow R_2$

$$\begin{bmatrix} 1 & -5 & 0 \\ 0 & 1 & -\frac{1}{9} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{2}{3} \end{bmatrix}$$

No it is not a vector space because zero vector not in the space

$x = \frac{10}{3} + \frac{5q}{9}$

$x + 5y = 0$

$x = -5y$

$y - \frac{q}{9} = \frac{2}{3}$

$\frac{6+q}{9} = y = \frac{2}{3} + \frac{q}{9}$

$\frac{30+5q}{9} = x$

$r = 2$

$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ 4 \end{bmatrix} \right\} = C(A)$

2D

2D inside 2-D

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{30}{9} + \frac{5q}{9} \\ \frac{6}{9} + \frac{q}{9} \\ q \end{pmatrix}$$

$$q \begin{pmatrix} \frac{5}{9} \\ \frac{1}{9} \\ 1 \end{pmatrix} = N(A)$$

1D in 3-D

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Question3 –

a-)

$$H^2 = (I - 2uu^T)(I - 2uu^T)$$

$$= I^2 - I\,2uu^T - 2uu^T.I + 4uu^Tuu^T$$

$$= I(I - 2uu^T - 2uu^T) + 4u\|u\|^2.u^T$$

$$= I - 2uu^T - 2uu^T + 4u.\|u\|^2.u^T$$

$$\|u\|^2 = u^T.u$$

$$\|u\| = \pm\sqrt{u^T.u}$$

$$\|u\| = 1$$

$$H^2 = I - 4uu^T + 4uu^T$$

$$H^2 = I - 4(1) + 4(1)$$

$$\boxed{H^2 = I}$$

b) $H.H^T = (I - 2uu^T)(I - 2uu^T)^T$

$= (I - 2uu^T)(I^T - (2uu^T)^T)$

$= (I - 2uu^T) - (I - 2(u^T)^T u^T)$

$= I - 2uu^T)(I - 2uu^T)$

$= I^2 - 2uu^T - 2uu^T + 4uu^T.uu^T$

$= I - 4uu^T + 4u\|u\|^2 u^T$

$= I - 4(1) + 4(1)$

$H.H^T = I \qquad H.H^T = H^T.H$ it is an Orthogonal Matrix

$H^T.H = I$

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Question 3 -

c) eigenvalue λ if $\boxed{Hv = \lambda v}$

$Hv = (I - 2uu^T).v$

$= Iv - 2uu^Tv$

$= v - 2u\|u\|^2$

eigenvalue $I = 1$

$Iv = 1.v = v$ and $\|u\| = \pm\sqrt{u^T.u}$

$u^Tu = \|u\|^2 = 1$

$Hv = v - 2u.1 = -u$

$Hu = -u = (-1).u$

$Hu = \lambda u$ $\boxed{\lambda = -1}$

u is an eigenvector of H

corresponding eigenvalue for u is $= \lambda = -1$

d) $H = (I - 2uu^T)$

$H^T = [I - 2uu^T]^T$

$= I^T - 2(uu^T)^T$ $\quad (A - B)^T = A^T - B^T$

$= I - 2(u^T)^T u^T$ $\quad (AB)^T = B^T A^T$

$= I = -2uu^T = H$

H is a symetric matrix

$H = Q \Lambda Q^T$ (where Q is an orthogonal) $\quad$ it is diagonalizable matrix

$= Q \Lambda Q^{-1}$

$= (Q^{-1})^{-1} \Lambda Q^{-1}$ $\quad [As,\ Q Q^T = I \ \ Q^T = Q^{-1}]$

$= P^{-1} \Lambda P$

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Question 4-

a-) first matrix → it is not a differential, it is a discrete. So we cannot use this matrix,

third matrix → it is not available to use condition

second matrix → it is a dif. equ. and available to use condition.

$$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$$

b-)

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Question 4-)

a)
$$\begin{bmatrix} -1-\lambda & 1 \\ 1 & -1-\lambda \end{bmatrix}$$

$$(-1-\lambda)(-1-\lambda)-1$$

$$1+2\lambda+(\lambda)^2-1=0$$

$$\lambda^2+2\lambda=0$$

$$\boxed{\lambda=0} \quad \boxed{\lambda=-2}$$

$$\lambda_0 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\boxed{X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

$$\lambda_{-2} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$v(t) = S_1 e^{x_1 t} c_1 + S_2 e^{x_2 t} c_2$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{0} c_1 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t} c_2$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} c_1 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t} c_2$$

$$0$$

$c_1 -$

$t \to \infty$

$\lim_{t \to \infty} \begin{bmatrix} 1 \\ 1 \end{bmatrix} < \infty$

$v_1 = v_2$

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